



NPTEL Course on “Computer System Optimization for ML”

Lecture #8

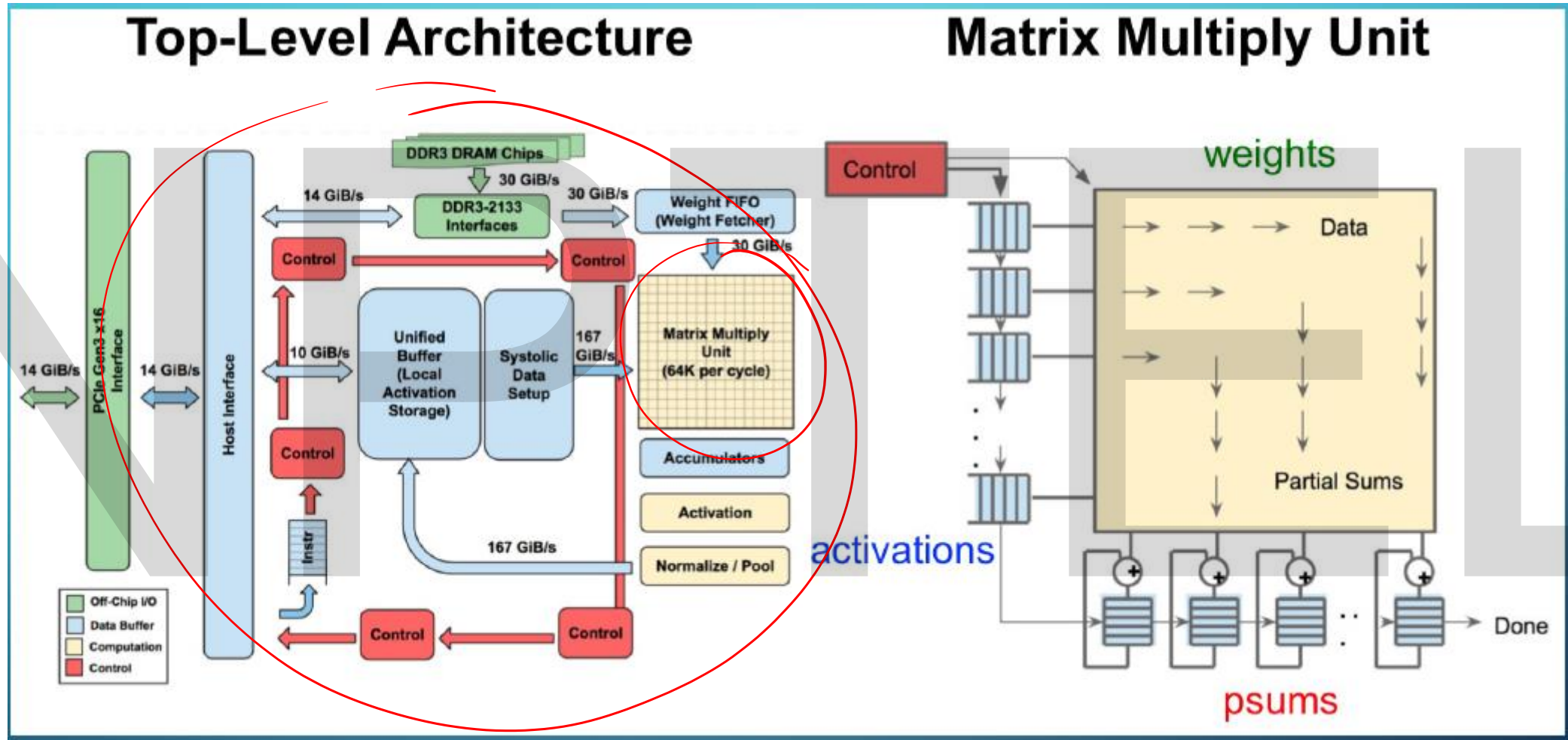


Agenda

- **Recap**
- **More Non-Linear Operations**
- **Popular CNNs**

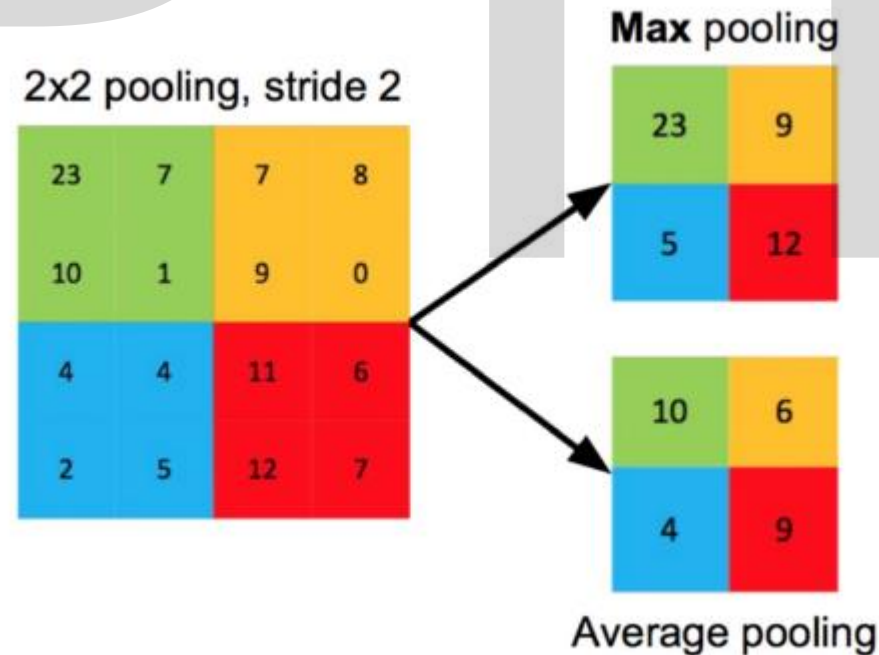
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Tensor Processing Unit



Pooling (Pool) Layer

- Reduce resolution of each channel independently
- Overlapping or non-overlapping
 - Depends on stride
- Increases translational-invariance and noise-resillience



Agenda

- Recap
- **More Non-Linear Operations**
- Popular CNNs

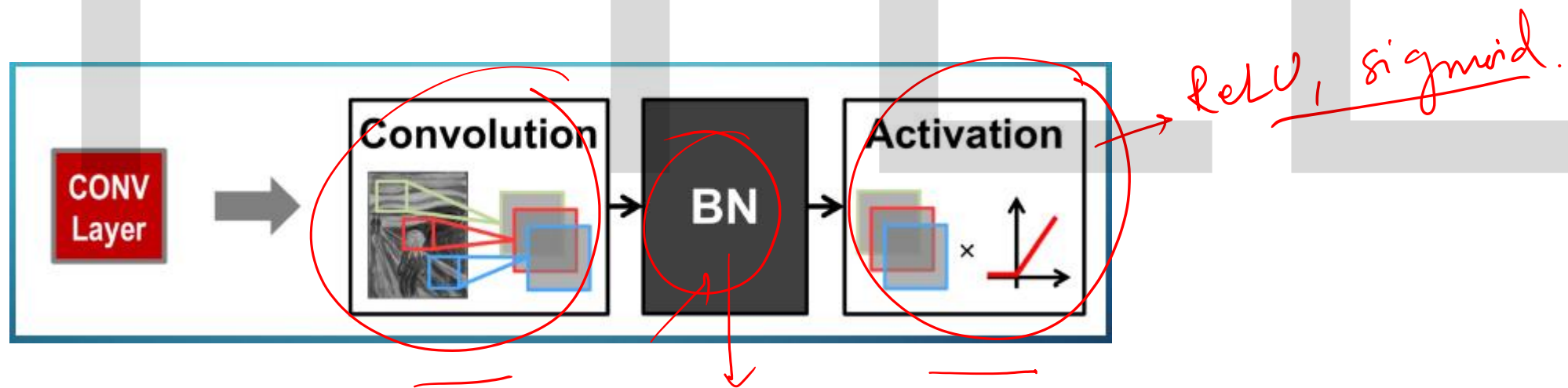
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Normalization Layer

- **Batch Normalization**

- Normalization activations towards mean=0 and std dev=1 based on the statistics of the training data set
- Put between conv/FC and activation function

- **Believed to be key to getting high accuracy and faster training for DNNs**



Normalization Layer

- The normalized values are further scaled and shifted
 - The parameters are learnt through training

The diagram shows the normalization equation $y = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \gamma + \beta$ with several annotations:

- data mean**: A red arrow points to μ , which is circled in red.
- learned scale factor**: A blue arrow points to γ , which is circled in red.
- data std. dev.**: A red arrow points to σ^2 , which is circled in red.
- learned shift factor**: A blue arrow points to β , which is circled in red.
- small const. to avoid numerical problems**: A red arrow points to ϵ , which is circled in red.
- A handwritten red note *mean* with a slash is next to the y variable.

The diagram illustrates a deep learning architecture for image classification. It starts with an input image (The Scream) which is processed by a **CONV Layer** to produce **Low-Level Features**. These features are then passed through another **CONV Layer** to produce **High-Level Features**. The high-level features are then processed by a **FC Layer** (Fully Connected Layer) to produce the final **Classes**. A detailed view of the **Convolution** process shows an input image being convolved with a kernel to produce a feature map, which is then passed through an **Activation** function (represented by a ReLU graph). The final output is a probability distribution over classes, with the most likely class (6) highlighted.

Agenda

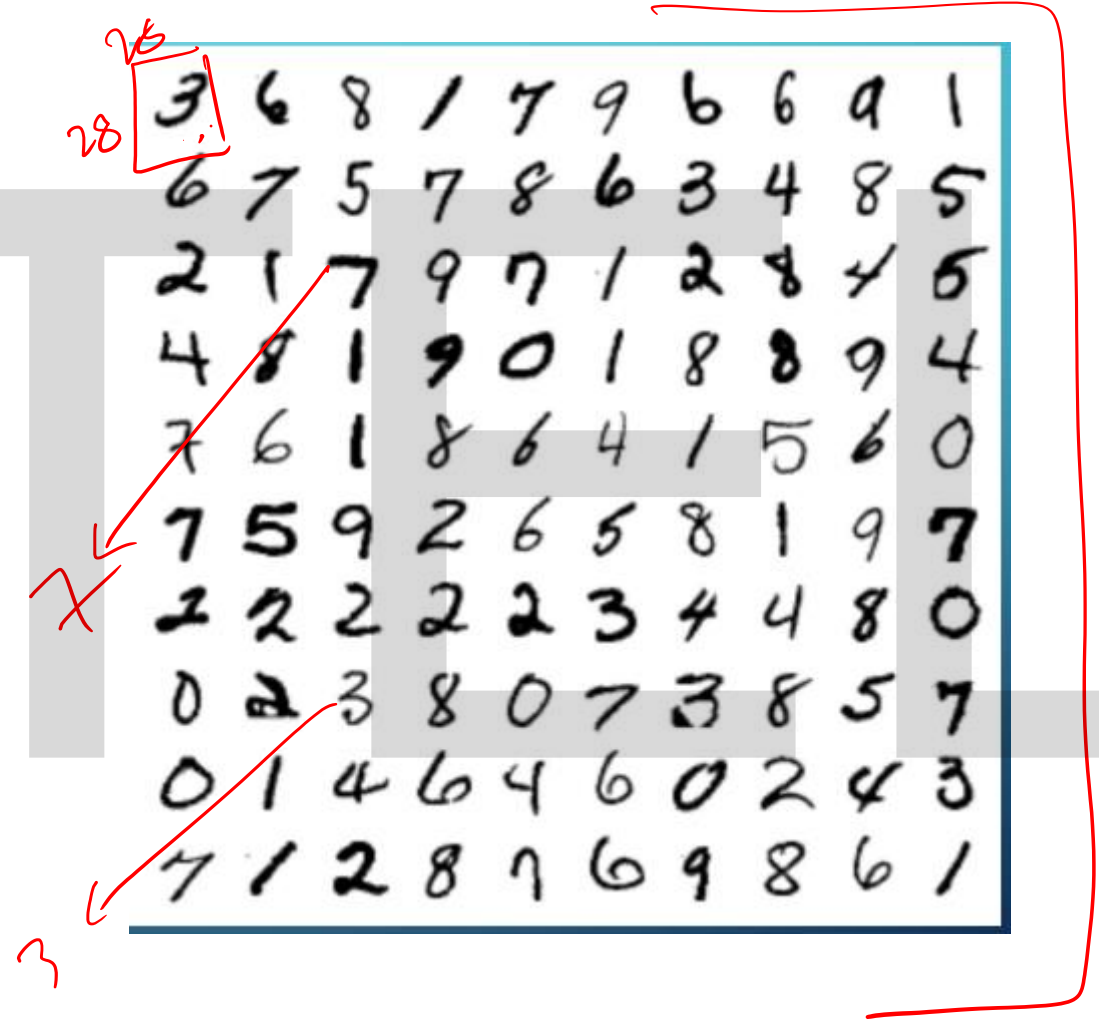
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Commonly used Dataset

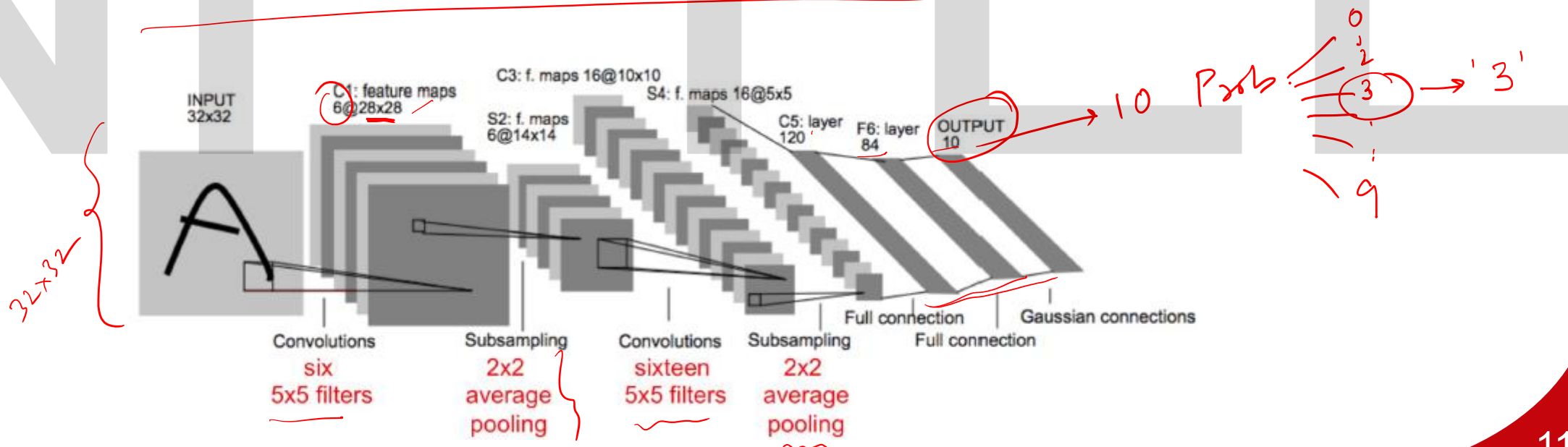
- **MNIST**

- Digit Classification
- 28×28 pixels (B&W)
- 10 Classes
- 60,000 training
- 10,000 testing

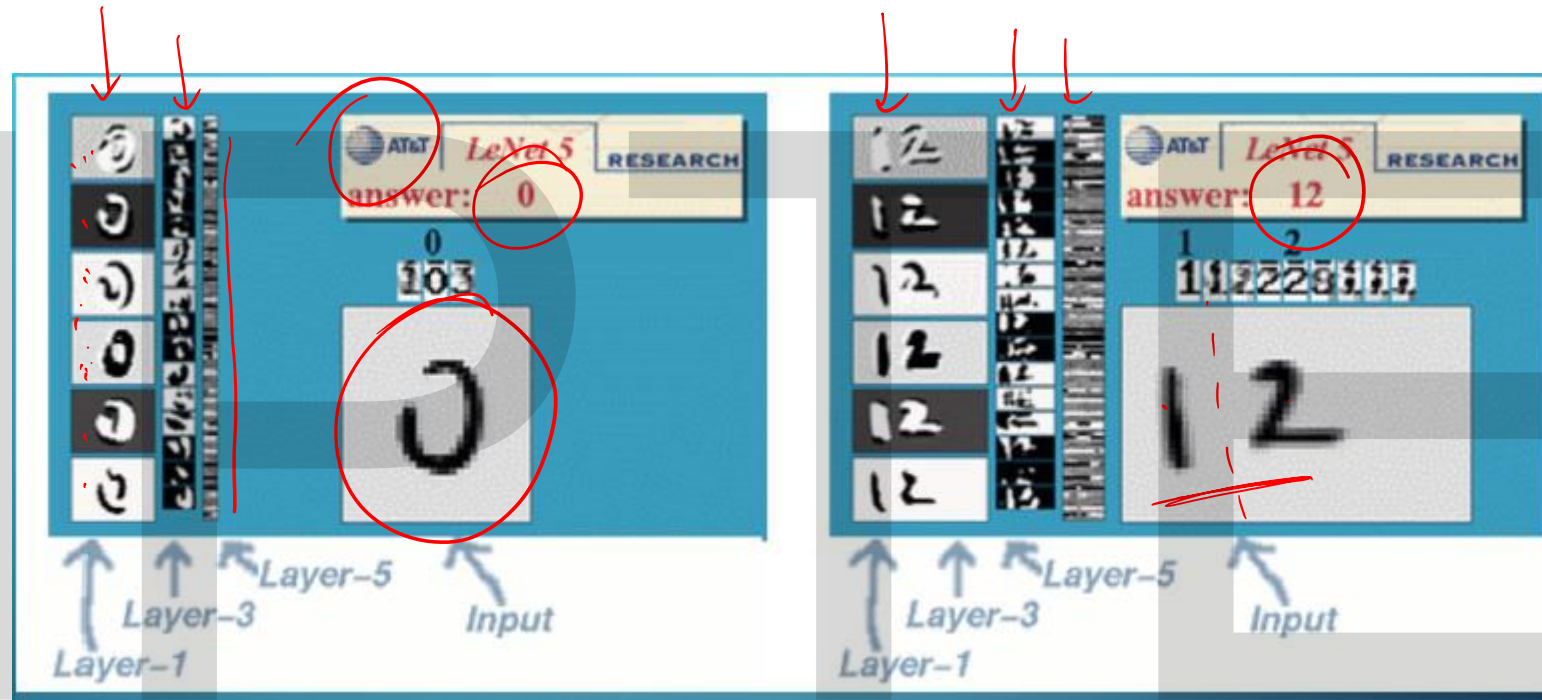


LeNet-5

- Conv layers: 2
- Fully connected layers: 2
- Weights: 60k
- MACs: 341k
- Sigmoid used for non-linearity



LeNet-5



ImageNet

- **Image classification**

– 256×256

- Colour images

– 1000 classes

– 1.3 M training

~~-100,000~~ testing

- **For ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**

- Accuracy of classification task reported based on top-1/top-5 error



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ImageNet

- **Image classification**

- 256×256
- Colour images
- 1000 classes
- 1.3 M training
- 100,000 testing

- **For ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**

- Accuracy of classification task reported based on top-1/top-5 error
- What is top-K error?



AlexNet (Krizhevsky et al., NeurIPS 2012)

- ILSCVR12 Winner
- Uses local response normalization (LRN)
- **Structure**
 - 5 conv layers
 - 3 fully connected layers
 - Weights: 61M
 - MACs 724M
 - ReLU used for non-linearity

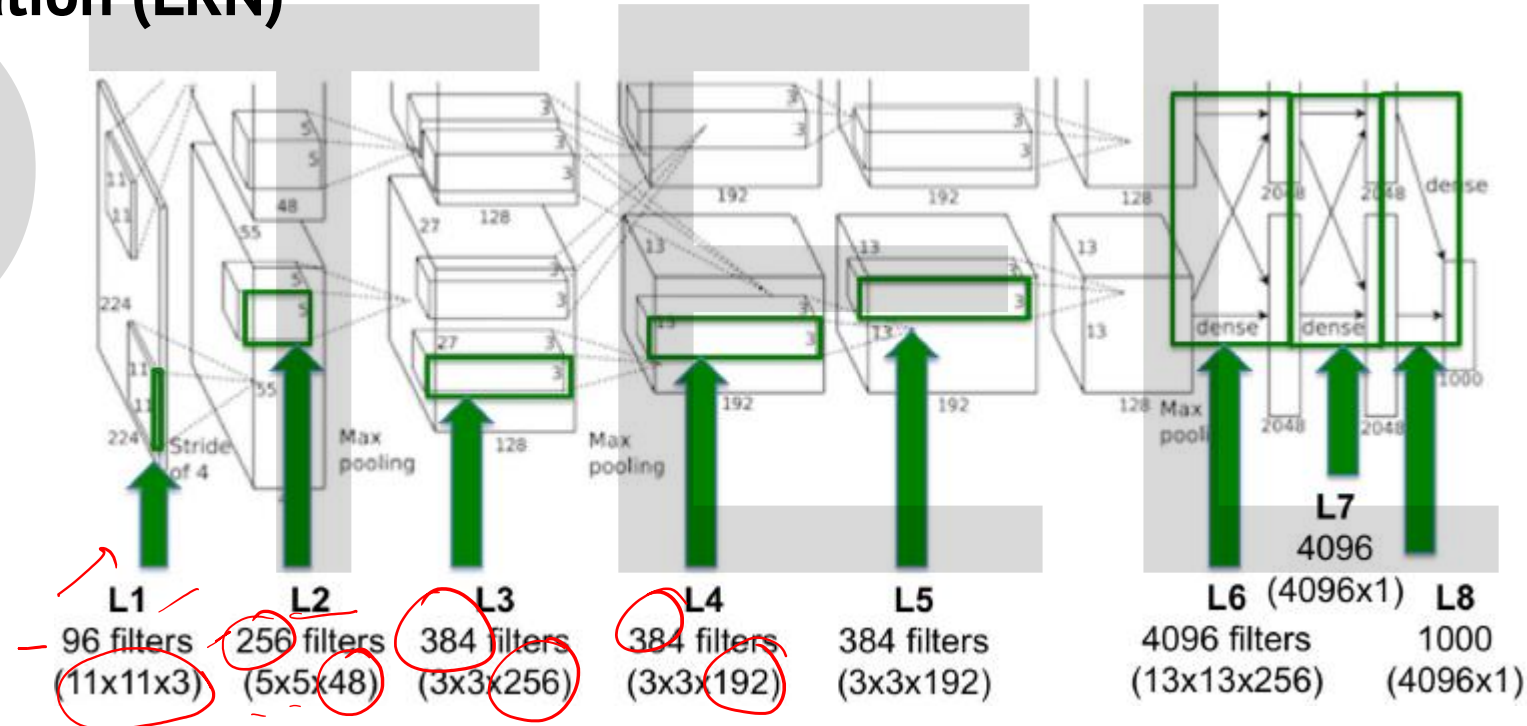


AlexNet (Krizhevsky et al., NeurIPS 2012)

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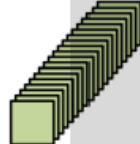
- **Structure**

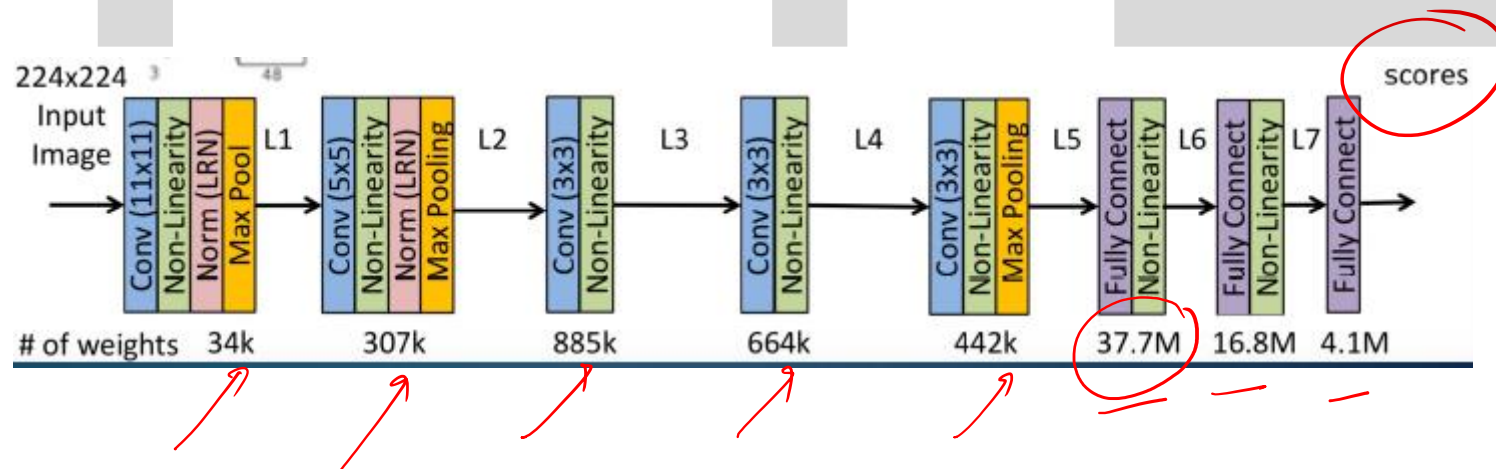
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AlexNet: Large Sizes with Varying Shapes

Layer	Filter Size (R x S)	# Filters (M)	# Channels (C)	Stride
1	11x11	96	3	4
2	5x5	256	48	1
3	3x3	384	256	1
4	3x3	384	192	1
5	3x3	256	192	1

Layer 1	Layer 2	Layer 3
		
34k Params 105M MACs	307k Params 224M MACs	885k Params 150M MACs



VGG-16 (Simonyan et al., ICLR 2015)

- Conv layers: 13
- FC layers: 3
- Weights: 138M
- MACs: 15.5G
- There is a 19-layer version too (VGG-19)

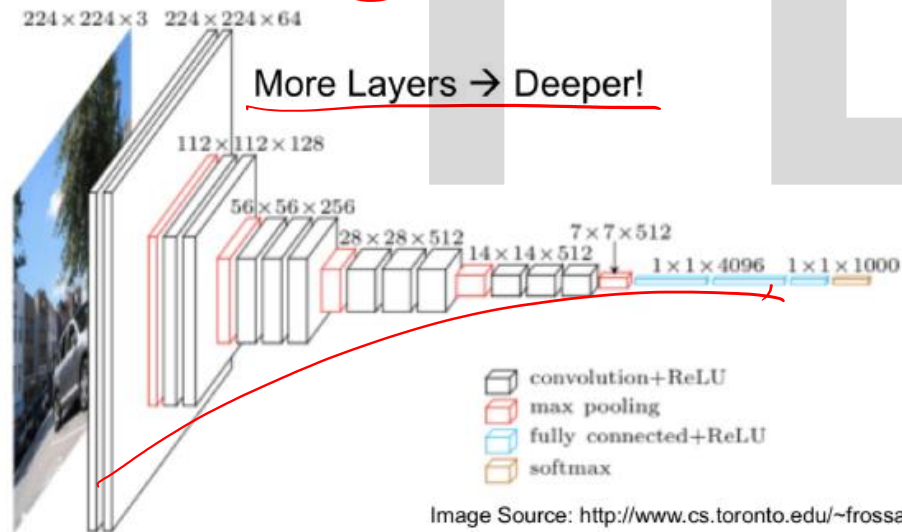
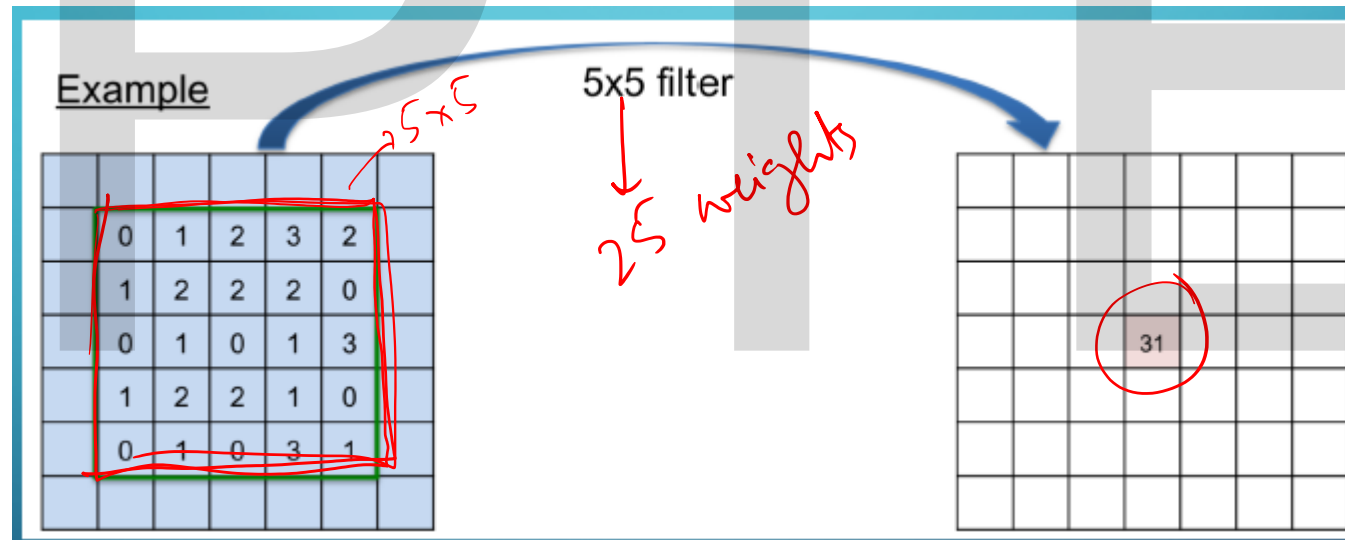


Image Source: <http://www.cs.toronto.edu/~frossard/post/vgg16/>

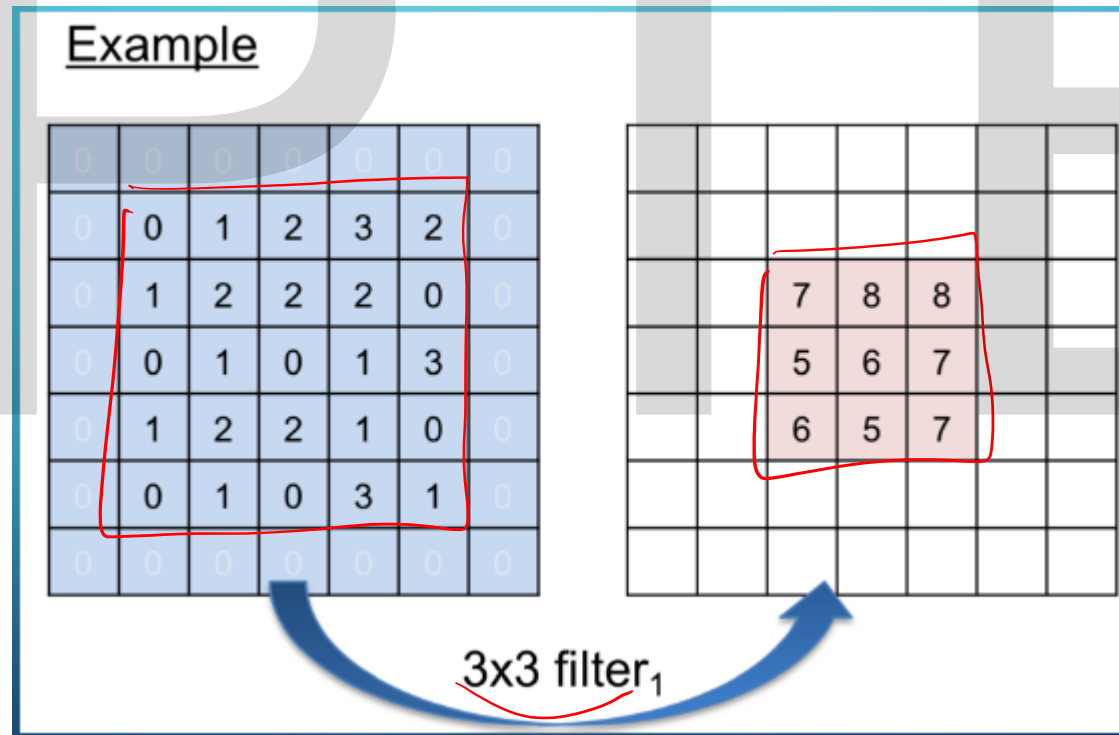
Stacked Filters

- Deeper networks means more weights
- Use stack of smaller filters (3×3) to cover the same receptive field with fewer filter weights



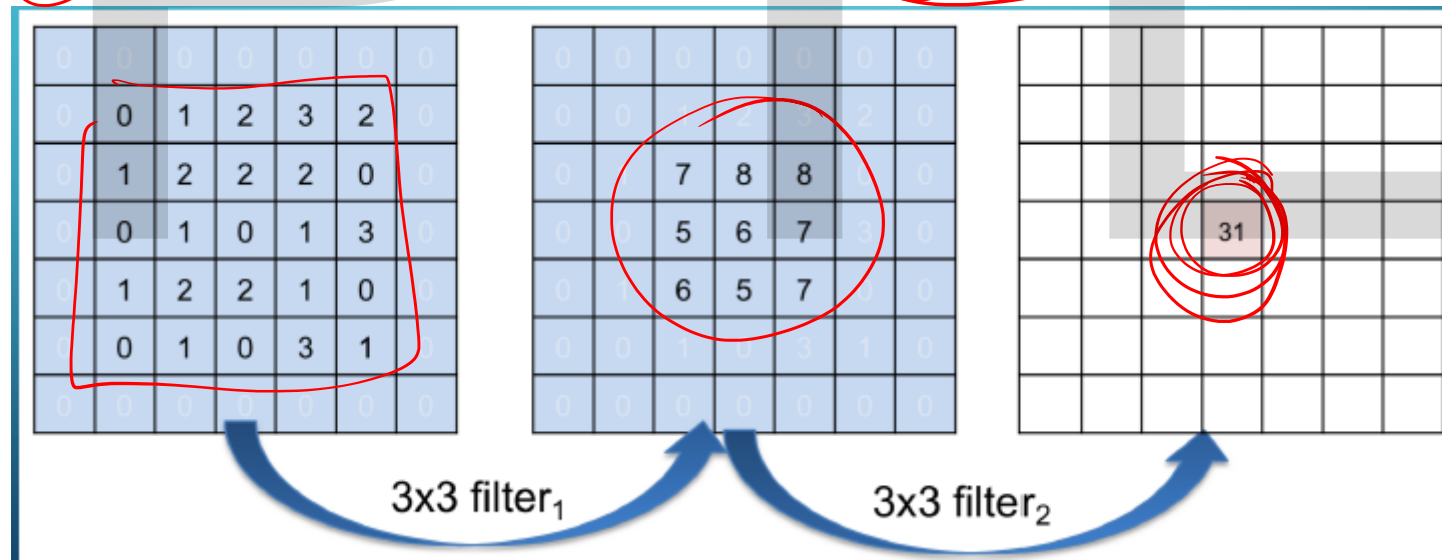
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Stacked Filters

- Deeper networks means more weights
- Use stack of smaller filters (3×3) to cover the same receptive field with fewer filter weights
- Non-linear activations inserted between each filter
 - 5×5 filter (25 weights) → two 3×3 filters (18 weights)





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)

